Dhanush Mathivanan

Data Scientist | Machine Learning Engineer | M.S. Data Science (May 2027)
Arlington, VA • 240-385-2816 • dhanush.mathivanan@gwu.edu • linkedin.com/in/dhanush-mathivanan
github.com/DhanushGWU1995 • Portfolio • techwithdhanush.blogspot.com

Professional Summary

Data scientist and machine learning engineer finishing an M.S. in Data Science at The George Washington University (May 2027), with 9.8+ years of software and data engineering experience on AWS. Recent work includes a grounded RAG assistant for technical manuals, SMS spam classifiers with DistilBERT and LIME (held-out Macro-F1 0.983), an LSTM complaint-prioritization system, and deep learning for facial pain detection. Before graduate school, built production data pipelines and APIs for Porsche, Educational Testing Service, and Roche. AWS Certified Solutions Architect and Developer (Associate). Seeking Data Scientist and Machine Learning Engineer roles starting May 2027.

Technical Skills

Languages: Python, SQL, R, TypeScript, JavaScript, Linux
Machine Learning & AI: Supervised learning, deep learning, NLP, generative AI, LLMs, RAG, embeddings, hybrid retrieval (BM25 + dense), computer vision, transfer learning, CNNs, LSTMs, feature engineering, model evaluation (precision, recall, F1, MRR)
ML Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, CatBoost, Hugging Face Transformers, sentence-transformers, Amazon Bedrock, Amazon Q, Amazon Lex
Data Analysis & Visualization: EDA, statistical analysis, Pandas, Polars, NumPy, DuckDB, PySpark, Jupyter, Tableau, Matplotlib, Seaborn, Streamlit
Data & Cloud: ETL/ELT, event-driven pipelines, Kafka, AWS EMR; AWS (S3, EC2, Lambda, API Gateway, DynamoDB); FastAPI, Docker, GitHub Actions

Education

The George Washington University, Washington, DC — Aug 2025 – May 2027
Master of Science in Data Science
Coursework: Big Data Analytics, Cloud Computing, Machine Learning, Deep Learning, NLP, Data Warehousing

Anna University, Tiruchirappalli, India — Aug 2012 – Apr 2016
Bachelor of Engineering in Computer Science
Coursework: Data Structures, Computer Programming, Software Engineering, Web Application Development, Algorithm Design

Projects

- **Torque Diagnostics (RAG, NLP):** Technician assistant for ABB drive manuals. Ingests PDFs to S3, prepares passages with Polars and DuckDB, retrieves with hybrid BM25 + BGE, drafts with Qwen2.5, and uses a DeBERTa gate to cite indexed pages or withhold unsupported procedures. Streamlit chat UI on GPU EC2; evaluated with precision, recall, F1, and MRR.
- **SMS Spam Detection / ScamShield (NLP, ML):** Classifiers for SMS spam and consumer-fraud education with TF-IDF logistic regression, DistilBERT, hybrid BM25 + FAISS retrieval, and LIME explanations; FastAPI and Angular UI. Hybrid NLP held-out Macro-F1 0.983 / accuracy 0.992.
- **ECRIS (NLP):** Complaint intelligence on CFPB consumer-finance data. An LSTM scores customer tone and urgency so complaints can be prioritized; live on Hugging Face Spaces (urgency F1 $\approx$ 0.94, tone F1 $\approx$ 0.88).
- **Facial Pain Detection (Deep Learning):** PyTorch models classify facial images into four pain levels, comparing a custom CNN with transfer learning on VGG-16, ResNet-50, and EfficientNet-B3; Grad-CAM explains each prediction.
- **Heart Disease Prediction (Machine Learning):** Cleaned and modeled CDC BRFSS survey data to predict coronary heart disease risk, and published a scoring web app.

Certifications & Awards

- AWS Certified Solutions Architect – Associate (SAA-C03), Amazon Web Services, 2024
- AWS Certified Developer – Associate (DVA-C02), Amazon Web Services, 2023
- Phi Delta Gamma Fellowship, The George Washington University, 2026
- Global Leaders Fellowship, The George Washington University, 2025
- Introduction to R, DataCamp, 2025
- Client Value Creation Award and Best People Award, Accenture, 2022

Professional Experience

The George Washington University, Washington, DC
Technical Support Assistant Aug 2026 – Present

- Cut estimated Tier 1 manual escalations by 42% by building an Amazon Connect, Lex, and Lambda contact center, then grounding staff answers with Bedrock and Amazon Q over internal docs.
- Built an EventBridge, S3, Bedrock, and Amazon Q pipeline that publishes support knowledge updates every 15 minutes and lets staff search internal documentation.
- Owned reliability for 10,000+ users and 5,000+ network endpoints on Cisco CUCM and RedSky, and moved incident logs into Splunk Cloud to speed root-cause analysis.

Probability Solutions LLC, Virginia
Data Science Intern Jun 2026 – Aug 2026

- Researched how binary search trees and related index structures behave in database engines at scale.
- Built Python prototypes and used statistical analysis to compare search, insert, update, and delete cost across tree implementations on large inputs.

Bosch Global Software Technologies, Client: MHP Porsche (Audi AG), Germany
Technical Project Specialist / Lead Engineer Mar 2024 – Aug 2025

- Shipped Porsche's workforce-planning platform on AWS serverless microservices (Lambda, API Gateway, DynamoDB, Cognito) with infrastructure as code in CDKTF and CloudFormation.
- Built S3-triggered Lambda ingestion with JSON schema validation (Joi) so enterprise HR files could be processed in real time.
- Kept legacy HR systems in sync through SNS, SQS, and Lambda event orchestration.

Accenture, Client: Educational Testing Service (ETS), USA
Custom Software Engineering Team Lead Sep 2021 – Mar 2024

- Cut exam-result processing time by 50% by redesigning scoring as serverless AWS services (Lambda, DynamoDB, SNS/SQS, CloudFormation) for a platform handling millions of exam records a year.
- Validated candidate score events with AJV and processed them through SQS-triggered Lambda workflows to keep high-volume runs stable.
- Automated DynamoDB maintenance in Python and Node.js and shipped reusable TypeScript service components across the platform APIs.

HCL Technologies, Client: Roche Holding AG, Switzerland
Lead Engineer – Full Stack Cloud Development Nov 2020 – Sep 2021

- Made diagnostic reporting workflows 40% faster with REST APIs and reporting services in Node.js, Express, Angular, and MySQL.
- Replaced manual lab reporting with automatically generated PDF and Excel reports from streamed data.
- Secured clinical integrations with OAuth/JWT APIs and AWS S3 storage for medical data exchange.

Object Frontier Software, Client: International Flavors & Fragrances (IFF), USA
Software Engineer – Full Stack Development Aug 2018 – Nov 2020

- Built an offline-first Electron desktop app with an embedded Node.js server and SQLite/MySQL sync for manufacturing and supply-chain work.
- Implemented Angular NgRx state management, including undo/redo across enterprise transactions.
- Connected Node.js microservices to MySQL with Sequelize and added structured logging with Winston.

V2Soft, Client: Fiat Chrysler Automobiles (FCA), USA
Software Engineer – HMI Software Development Oct 2016 – Aug 2018

- Built automotive HMI dashboards in Angular, HTML5, and Bootstrap for infotainment and diagnostics.
- Processed vehicle telemetry with MongoDB batch jobs and Node.js APIs.
- Connected dashboards to backend services with Socket.IO for live operational data.